

EDSON FUENTES

458-285-6851 | fuenteed@oregonstate.edu | [linkedin.com/in/edsonfuentes](https://www.linkedin.com/in/edsonfuentes) | github.com/fuenteed

Education

Oregon State University

Master of Engineering, Computer Science

Expected Summer 2027

Corvallis, OR

Oregon State University

B.S. Computer Science | Minor: Statistics | GPA: 3.7

Sep 2022 – Jun 2025

Corvallis, OR

Technical Skills

Languages: C, C++, Python, Rust, Bash

Embedded: STM32 / ARM Cortex-M, Zephyr RTOS (devicetree, Kconfig, west), Embedded Linux (Raspberry Pi CM5)

Interfaces: CAN / FDCAN, SocketCAN, CCP, SPI, I2C, UART, PWM, ADC

Tools: GDB, ST-Link, Vector CANoe, Twister, Docker, PlatformIO, CMake, Make, Git, oscilloscope

Experience

Ampere Computing

Jun 2026 – September 2026

System Control Firmware Intern

Portland, OR

- Owned migration of system control firmware from **Zephyr RTOS** v4.3 to v4.4, resolving **5** breaking upstream API and configuration changes and validating boot and runtime behavior on target hardware.
- Updated and debugged the **AHB** controller and **PL011 UART** Zephyr drivers in **C** for the v4.4 release.
- Validated the upgrade with **Twister** test suites in **Docker**-based CI on simulated targets, then on real SoC hardware in a remote lab.

Global Formula Racing (Oregon State University)

Feb 2026 – Present

Embedded Software Lead

Aug 2026 – Present

- Lead a **3**-person embedded team and own firmware for all **5+** vehicle electronics boards; split board ownership across members and wrote onboarding documentation for the firmware workflow.
- Leading development of a **C** telemetry system on a **Raspberry Pi CM5** that reads vehicle CAN traffic through SPI CAN controllers via **SocketCAN**, logs to onboard storage, and streams live data to a ground station.

Software Engineer

Feb 2026 – Aug 2026

- Developed **STM32G431** (Cortex-M4F) firmware in **C++** for a custom sensor/actuator PCB, generating PWM servo control and acquiring analog sensor data over the vehicle **CAN** network.
- Implemented a **CCP** slave enabling live calibration of servo pulse widths and sensor scaling from **Vector CANoe** without reflashing; verified with CAN traces and oscilloscope captures.

Oregon State University

Sep 2025 – Present

Graduate Teaching Assistant, Statistics

Corvallis, OR

- Support **180+** students in statistical inference, regression, and ANOVA, diagnosing misconceptions and grading **R**-based assignments with actionable feedback.

Projects

Small Shell in C | C, POSIX, Signals, Process Control, Memory Management

Oct 2022

- Built a Unix-like shell in **C** supporting built-ins, I/O redirection, and external command execution via **fork/exec** and **waitpid**.
- Implemented **SIGINT/SIGTSTP** handling for foreground/background process control; verified resource cleanup and memory correctness using **GDB**.

Leadership

Embedded Systems Design Club

Jun 2026 – Present

President

Oregon State University

- Planned a full academic year of club meetings, including **5** fall-term technical workshops on bare-metal C, peripheral interfacing, and hardware debugging on ARM Cortex-M.

Society of Hispanic Professional Engineers (SHPE)

May 2022 – Present

Member

Oregon State University